

Research Idea: SUNRISE_ARC

ALMA Thesis Planner (automated pipeline)

Title: The [CII]158 μ m arc at $z=5.93$: integrated FIR properties and an SNR-matched velocity cut along the 16'' lensed arc hosting Earendel

One-sentence version. Using only the ALMA program's own data (ADS/JAO.ALMA#2023.1.00802.S), the student delivers the system's first integrated FIR characterization — [CII] flux (or limit), line width, and velocity centroid for the full $\sim 16''$ arc, plus a dust-continuum flux or limit — and then extracts a position-velocity cut along the arc's ridge line at a spatial sampling *set by the achieved signal-to-noise rather than assumed*, asking whether the [CII]-emitting gas shows an ordered velocity gradient, distinct velocity components, or a single kinematically simple reservoir.

Week 0 gate (a few hours, before the project is approved). Query the ALMA Science Archive entry for 2023.1.00802.S: observation status, proprietary-expiry date, delivered beam size, spectral setup, and achieved RMS per channel from the archive/QA2 products. Every number that later sizing depends on comes from here, never from the requested (as-proposed) configuration. Three quantities are written down in week 0 and fixed:

- σ_{chan} , the achieved RMS per channel per beam;
- N_{beam} , the number of independent beams along the 16'' arc;
- the resulting **per-beam velocity-centroid threshold**. A velocity centroid is only quoted where the line is detected at $\geq 5\sigma$ in the integrated moment-0 within that aperture; centroid uncertainty scales roughly as $\text{FWHM}/(2.35 \cdot \text{SNR})$, so the student computes, in week 0, the SNR needed for a centroid precision meaningful against a plausible dwarf-galaxy line width (tens of km/s) and reports how many independent positions along the arc actually clear it.

Pre-specified SNR ladder for the core measurement (this replaces the assertion that per-beam velocities are free). Step 3 below is executed at whichever rung the week-0 arithmetic supports, and the rung is declared before the data are looked at:

- **Rung A** — per-beam spectra along the ridge, full PV diagram, per-beam centroid and line-width profiles.
- **Rung B** — centroids in N coarse bins along the arc, N chosen so each bin reaches the threshold SNR (e.g. 3–5 bins instead of ~ 15 beams). A three-bin blue/middle/red measurement across a 16'' arc is still a real kinematic result and is the *expected* outcome if the integrated line lands near the proposal's design sensitivity.
- **Rung C** — integrated line only, plus a red-half/blue-half velocity difference and a stacked-spectrum test for line asymmetry or a second component. This is a one-number kinematic result, but it is a measurement, not a null.

There is no rung at which step 3 produces nothing, and no rung is discovered in month six.

Why the spectral axis is the right core. The proposal’s own goal is the *initial* FIR characterization of this system. The integrated quantities are guaranteed to exist as results — a detection gives fluxes and a line profile, a non-detection gives limits — and either is a real, quotable measurement of a $z \sim 6$ dwarf that nobody has yet. The velocity axis is where the arc’s 16” extent pays off at compact-configuration resolution, because velocity information does not require resolving the clumps; but how *finely* it pays off is an empirical question answered by the rung ladder above, not an assumption.

Data and inputs (nothing beyond what the description names).

- ALMA [CII]158 μ m line and dust continuum for this target, compact + extended configurations, from the program named in the acknowledgement.
- The NIRSpc-confirmed redshift $z=5.93$, the $\sim 16''$ arc morphology, and Earendel itself, from the abstract and Welch+22 (Nature).
- **Named-input honesty.** The description mentions NIRCам observations mapping diffuse gas and several compact clumps, but names *no* paper or catalog tabulating clump coordinates — Welch+22 is cited only for Earendel. The entire thesis is therefore built to require no clump position list. The arc ridge line is traced from the ALMA continuum/[CII] emission itself together with the arc geometry described in Welch+22. If a clump coordinate list happens to become available from the collaboration during the year, the clump comparison is added as a bonus appendix; it is never scheduled, and nothing depends on it.

Frequency and measurement provenance (settled before any imaging, using only in-hand data). The description labels the line “[CII]158 μ m” but does not state whether that 158 μ m is rest-frame or observed-frame, and quotes no frequency at all — the student carries it as an unverified wavelength label and says so explicitly in the thesis. The expected observed-frame frequency is derived from the NIRSpc-confirmed $z=5.93$ quoted in the description and then checked **against the spectral-window setup recorded in the archived measurement sets**, which is self-contained evidence for where the observers actually placed the line and which channels are line-free. No external line database is consulted; none is named in the description and none is needed, because the correlator setup is the authoritative record for this dataset. The redshift is carried as “ $z=5.93$, NIRSpc-confirmed,” not as an exact adopted constant, and the line-search window is kept wide enough to allow a velocity offset between [CII] and the rest-UV/optical features that set that redshift. The magnification is quoted in the description as an approximate $\mu \sim 300$ for Earendel specifically — a highly localized value near the critical curve — and is never applied to the arc as a whole. The clump radii “between 1 and 20 pc” are NIRCам-inferred source-plane sizes and are carried as such, never as ALMA-measured sizes.

Work plan (one semester to one year).

1. *CASA training and reduction, weeks 1–10.* Budgeted for a student who is not yet fluent in CASA: weeks 1–3 are explicit training on the delivered calibrated products and the ALMA pipeline weblog before any science imaging. Then restore the archival calibrated data, inspect flagging and calibrator solutions, and produce a continuum image from line-free channels and a [CII] cube from the compact configuration at native resolution and one taper. If the calibrated-products path (using delivered pipeline products rather than a full re-restore) is available from the archive, take it and say so — the thesis’s value is not in re-deriving calibration. Compact configuration is the primary dataset throughout. 2. *Integrated [CII] and continuum, weeks 10–15 — result #1.* Aperture-integrated [CII] spectrum over the whole arc: flux (or limit), FWHM, and

velocity centroid relative to the NIRSpec redshift; dust-continuum flux or upper limit; apparent [CII] luminosity and apparent L_{IR} . Self-contained regardless of everything that follows. **The measured integrated SNR here confirms or demotes the week-0 rung choice**, and the rung is locked at this point. 3. *Position-velocity cut along the arc, weeks 15–26 — the thesis core, executed at the locked rung.* Trace the ridge line, extract spectra at the rung-appropriate sampling, and build the PV diagram (Rung A/B) or the halves comparison (Rung C), with per-position velocity centroid and line-width profiles. All uncertainties come from identical extractions at identical apertures on emission-free sky, not from formal fit errors alone. Deliverables: is there a monotonic velocity gradient along the arc; are there distinct velocity components; does the line width vary systematically? Each is reported with confidence intervals and tested against a flat-velocity null. 4. *Interpretation with the lensing caveat carried honestly, weeks 26–29.* No lens model is named in the description, so all results are stated as **image-plane** observables. The thesis does not claim that smooth image-plane emission implies diffuse source-plane gas, nor that an image-plane velocity gradient implies source-plane rotation — differential magnification along a fold can smooth genuinely clumpy emission, and mirrored images can mimic or erase an ordered gradient. Those inferences are explicitly withheld pending a lens model. What the image-plane result *does* support without a model is follow-up design: the observed surface-brightness and velocity contrast on the sky is exactly the quantity a future long-baseline program must detect, and the achieved per-beam, per-channel sensitivity converts directly into required integration times. That framing is magnification-agnostic and is stated as the payoff. 5. *Bounded final chapter — what these data can and cannot separate, weeks 29–34.* One injection-recovery campaign, run in the visibility domain on the real data at the true delivered beam and RMS, injecting a clump-following template (clumps placed along the traced ridge at a range of spacings and contrasts, since no coordinate list is assumed) and a constant-along-arc template. The output is a quantitative statement of the clump-to-diffuse contrast these data can and cannot distinguish, expressed as a design requirement — beam size, sensitivity, and configuration needed to separate clumps from diffuse gas at $z=6$. This is a chapter, not the thesis, and it yields a number either way. As a short exploratory add-on only, the 1D brightness and velocity profiles are searched for reflection symmetry about a free-position axis, reported strictly as a bound with the caveat stated plainly: fitting such an axis in a low-SNR 1D profile is weakly constrained, and the axis being fitted is effectively the critical-curve position, so this is not a model-free test and is not advertised as one.

Explicit scope cuts. The extended-configuration data are *not* imaged clump-by-clump; they are inspected only far enough to report whether a significant fraction of [CII] flux is missing on long baselines, and only if steps 1–3 finish on schedule. No separate simulated-cube campaign is run — there is exactly one injection campaign, on real visibilities. Clump-by-clump imaging is flagged in the thesis as the natural next project and as a deliberate scope choice, not a limitation of the data.

Proprietary-data risk, stated without a fake rescue. If week 0 shows the data are proprietary beyond the thesis window, there is no honest substitute: the data description names no other dataset, catalog, or archive for this target — only Welch+22 for Earendel and the NIRCам observations described without a citable data product — so this idea does not invent one. The fallback is stated plainly as *reduced in scope*, not equivalent: a methods-and-forecast thesis consisting of (i) the full extraction and injection-recovery pipeline built and validated on CASA-simulated cubes matched to the as-proposed configurations, (ii) an explicit sensitivity forecast for what the delivered data will be able to measure, and (iii) a literature synthesis of published work on this arc. That is a defensible one-semester project and a thin one-year project, and the right response to a proprietary result at week 0 is for the advisor to pair it with a second project from outside this dataset — which is a supervision decision, not something this idea can manufacture from the

description in front of it.

Assumptions stated, not hidden. All luminosities and star-formation rates are reported as apparent, magnification-scaled quantities ($\mu \cdot L_{\text{[CII]}}$, $\mu \cdot L_{\text{IR}}$, $\mu \cdot \text{SFR}$), with intrinsic values deferred and flagged as requiring a lens model outside this dataset. Any conversion of continuum flux to an obscured SFR requires an assumed dust temperature and emissivity index, so results are quoted over a stated grid of assumptions — a single-band continuum measurement cannot constrain the dust SED.

Deliverables. A reduced [CII] cube and continuum image; an integrated [CII] flux (or limit), FWHM, and velocity centroid for the $\sim 16''$ arc referenced to the NIRSpec-confirmed $z=5.93$; a dust-continuum flux or limit and apparent L_{IR} over a stated dust-SED grid; a kinematic profile along the arc ridge at the SNR-justified sampling (PV diagram, coarse-bin centroids, or halves comparison) with quantitative tests for a gradient and for multiple components; a measured recovered-flux sensitivity curve stating what clump-versus-diffuse contrast these data can distinguish; and an evidence-based specification of the resolution and sensitivity a follow-up high-resolution program would need.